MATH155: Polynomial and Rational Functions

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Name: ______________________________ Date: _______________ Score: / 30

Learning Objectives

- Solve quadratic equations by factoring and the quadratic formula
- Find the vertex of a parabola in standard form
- Identify excluded values from the domain of a rational function
- Determine horizontal and vertical asymptotes of rational functions

Simplify each expression completely. Show all steps and circle your final answer.

Solving quadratic equations

1. Solve the quadratic equation: 1x^2 + 1x + -20 = 0.

$$x^2 + 1x - 20 = 0$$

Answer: ____

2. Solve the quadratic equation: 2x^2 + 12x + 10 = 0.

$$2x^2 + 12x + 10 = 0$$

Answer: ____

3. Solve the quadratic equation: 2x^2 + 16x + 24 = 0.

$$2x^2 + 16x + 24 = 0$$

Answer: ____

4. Solve the quadratic equation: 1x^2 + 7x + 6 = 0.

$$x^2 + 7x + 6 = 0$$

Answer: ____

5. Solve the quadratic equation: 1x^2 + 1x + -12 = 0.

$$x^2 + 1x - 12 = 0$$

Answer: ____

6. Solve the quadratic equation: 2x^2 + 20x + 50 = 0.

$$2x^2 + 20x + 50 = 0$$

Answer: ____

Quadratic formula — discriminant

7. Use the quadratic formula to solve x^2 + 2x + -15 = 0 (a = 1, roots are integers).

$$x^2 + 2x - 15 = 0$$

Answer: ____

8. Use the quadratic formula to solve x^2 + -2x + -3 = 0 (a = 1, roots are integers).

$$x^2 - 2x - 3 = 0$$

Answer: ____

9. Use the quadratic formula to solve x^2 + -1x + -2 = 0 (a = 1, roots are integers).

$$x^2 - 1x - 2 = 0$$

Answer: ____

10. Use the quadratic formula to solve x^2 + -1x + -2 = 0 (a = 1, roots are integers).

$$x^2 - 1x - 2 = 0$$

Answer: ____

11. Use the quadratic formula to solve x^2 + -2x + -8 = 0 (a = 1, roots are integers).

$$x^2 - 2x - 8 = 0$$

Answer: ____

12. Use the quadratic formula to solve x^2 + -3x + -10 = 0 (a = 1, roots are integers).

$$x^2 - 3x - 10 = 0$$

Answer: ____

Vertex of a parabola

13. Find the vertex of the parabola f(x) = -3x^2 + 18x + -33.

$$f(x) = -3x^2 + 18x - 33$$

Answer: ____

14. Find the vertex of the parabola f(x) = 2x^2 + -4x + -4.

$$f(x) = 2x^2 - 4x - 4$$

Answer: ____

15. Find the vertex of the parabola f(x) = 2x^2 + -12x + 18.

$$f(x) = 2x^2 - 12x + 18$$

Answer: ____

16. Find the vertex of the parabola f(x) = 1x^2 + -10x + 30.

$$f(x) = 1x^2 - 10x + 30$$

Answer: ____

17. Find the vertex of the parabola f(x) = -1x^2 + 10x + -22.

$$f(x) = -1x^2 + 10x - 22$$

Answer: ____

18. Find the vertex of the parabola f(x) = 1x^2 + 6x + 2.

$$f(x) = 1x^2 + 6x + 2$$

Answer: ____

Domain of a rational function

19. Find the value(s) excluded from the domain of f(x) = (3x + 1) / (4x + -1).

$$f(x) = \frac{3x + 1}{4x + -1}$$

Answer: ____

20. Find the excluded value from the domain of g(x) = 2 / (x - 4).

$$g(x) = \frac{2}{x - 4}$$

Answer: ____

21. Find the value(s) excluded from the domain of f(x) = (4x + -1) / (3x + -2).

$$f(x) = \frac{4x + -1}{3x + -2}$$

Answer: ____

22. Find the excluded value from the domain of g(x) = 4 / (x - 8).

$$g(x) = \frac{4}{x - 8}$$

Answer: ____

23. Find the value(s) excluded from the domain of f(x) = (1x + -2) / (2x + 0).

$$f(x) = \frac{1x + -2}{2x + 0}$$

Answer: ____

24. Find the excluded value from the domain of g(x) = 9 / (x - -8).

$$g(x) = \frac{9}{x - -8}$$

Answer: ____

25. Find the value(s) excluded from the domain of f(x) = (3x + -1) / (5x + -5).

$$f(x) = \frac{3x + -1}{5x + -5}$$

Answer: ____

26. Find the excluded value from the domain of g(x) = 8 / (x - -3).

$$g(x) = \frac{8}{x - -3}$$

Answer: ____

27. Find the value(s) excluded from the domain of f(x) = (4x + -1) / (3x + 7).

$$f(x) = \frac{4x + -1}{3x + 7}$$

Answer: ____

28. Find the excluded value from the domain of g(x) = 2 / (x - 1).

$$g(x) = \frac{2}{x - 1}$$

Answer: ____

29. Find the value(s) excluded from the domain of f(x) = (1x + -1) / (4x + 5).

$$f(x) = \frac{1x + -1}{4x + 5}$$

Answer: ____

30. Find the excluded value from the domain of g(x) = 6 / (x - 8).

$$g(x) = \frac{6}{x - 8}$$

Answer: ____

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ANSWER KEY & SOLUTIONS

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Topics: Solving quadratic equations, Quadratic formula — discriminant, Domain of a rational function, Vertex of a parabola. All answers verified by independent computation.

Solutions

Solving quadratic equations

1. Solve the quadratic equation: 1x^2 + 1x + -20 = 0.

$$x^2 + 1x - 20 = 0$$

→ *The equation factors as 1(x - (4))(x - (-5)) = 0.*
→ *Set each factor to zero: x - (4) = 0 or x - (-5) = 0.*
→ *Solutions: x = -5 and x = 4.*

Answer: $x = -5$ or $x = 4$

2. Solve the quadratic equation: 2x^2 + 12x + 10 = 0.

$$2x^2 + 12x + 10 = 0$$

→ *The equation factors as 2(x - (-1))(x - (-5)) = 0.*
→ *Set each factor to zero: x - (-1) = 0 or x - (-5) = 0.*
→ *Solutions: x = -5 and x = -1.*

Answer: $x = -5$ or $x = -1$

3. Solve the quadratic equation: 2x^2 + 16x + 24 = 0.

$$2x^2 + 16x + 24 = 0$$

→ *The equation factors as 2(x - (-2))(x - (-6)) = 0.*
→ *Set each factor to zero: x - (-2) = 0 or x - (-6) = 0.*
→ *Solutions: x = -6 and x = -2.*

Answer: $x = -6$ or $x = -2$

4. Solve the quadratic equation: 1x^2 + 7x + 6 = 0.

$$x^2 + 7x + 6 = 0$$

→ *The equation factors as 1(x - (-6))(x - (-1)) = 0.*
→ *Set each factor to zero: x - (-6) = 0 or x - (-1) = 0.*
→ *Solutions: x = -6 and x = -1.*

Answer: $x = -6$ or $x = -1$

5. Solve the quadratic equation: 1x^2 + 1x + -12 = 0.

$$x^2 + 1x - 12 = 0$$

→ *The equation factors as 1(x - (3))(x - (-4)) = 0.*
→ *Set each factor to zero: x - (3) = 0 or x - (-4) = 0.*
→ *Solutions: x = -4 and x = 3.*

Answer: $x = -4$ or $x = 3$

6. Solve the quadratic equation: 2x^2 + 20x + 50 = 0.

$$2x^2 + 20x + 50 = 0$$

→ *The equation factors as 2(x - (-5))(x - (-5)) = 0.*
→ *Set each factor to zero: x - (-5) = 0 or x - (-5) = 0.*
→ *Solutions: x = -5 and x = -5.*

Answer: $x = -5$ or $x = -5$

Quadratic formula — discriminant

7. Use the quadratic formula to solve x^2 + 2x + -15 = 0 (a = 1, roots are integers).

$$x^2 + 2x - 15 = 0$$

→ *Identify: a = 1, b = 2, c = -15.*
→ *Discriminant: b^2 - 4ac = (2)^2 - 4(1)(-15) = 64.*
→ *x = (-2 +/- sqrt(64)) / 2 = (-2 +/- 8) / 2.*
→ *x = -5 or x = 3.*

Answer: $x = -5 \text{ or } x = 3$

8. Use the quadratic formula to solve x^2 + -2x + -3 = 0 (a = 1, roots are integers).

$$x^2 - 2x - 3 = 0$$

→ *Identify: a = 1, b = -2, c = -3.*
→ *Discriminant: b^2 - 4ac = (-2)^2 - 4(1)(-3) = 16.*
→ *x = (--2 +/- sqrt(16)) / 2 = (2 +/- 4) / 2.*
→ *x = -1 or x = 3.*

Answer: $x = -1 \text{ or } x = 3$

9. Use the quadratic formula to solve x^2 + -1x + -2 = 0 (a = 1, roots are integers).

$$x^2 - 1x - 2 = 0$$

→ *Identify: a = 1, b = -1, c = -2.*
→ *Discriminant: b^2 - 4ac = (-1)^2 - 4(1)(-2) = 9.*
→ *x = (--1 +/- sqrt(9)) / 2 = (1 +/- 3) / 2.*
→ *x = -1 or x = 2.*

Answer: $x = -1 \text{ or } x = 2$

10. Use the quadratic formula to solve x^2 + -1x + -2 = 0 (a = 1, roots are integers).

$$x^2 - 1x - 2 = 0$$

→ *Identify: a = 1, b = -1, c = -2.*
→ *Discriminant: b^2 - 4ac = (-1)^2 - 4(1)(-2) = 9.*
→ *x = (--1 +/- sqrt(9)) / 2 = (1 +/- 3) / 2.*
→ *x = -1 or x = 2.*

Answer: $x = -1 \text{ or } x = 2$

11. Use the quadratic formula to solve x^2 + -2x + -8 = 0 (a = 1, roots are integers).

$$x^2 - 2x - 8 = 0$$

→ *Identify: a = 1, b = -2, c = -8.*
→ *Discriminant: b^2 - 4ac = (-2)^2 - 4(1)(-8) = 36.*
→ *x = (--2 +/- sqrt(36)) / 2 = (2 +/- 6) / 2.*
→ *x = -2 or x = 4.*

Answer: $x = -2 \text{ or } x = 4$

12. Use the quadratic formula to solve x^2 + -3x + -10 = 0 (a = 1, roots are integers).

$$x^2 - 3x - 10 = 0$$

→ *Identify: a = 1, b = -3, c = -10.*
→ *Discriminant: b^2 - 4ac = (-3)^2 - 4(1)(-10) = 49.*
→ *x = (--3 +/- sqrt(49)) / 2 = (3 +/- 7) / 2.*
→ *x = -2 or x = 5.*

Answer: $x = -2$ or $x = 5$

Vertex of a parabola

13. Find the vertex of the parabola f(x) = -3x^2 + 18x + -33.

$$f(x) = -3x^2 + 18x - 33$$

→ *Use the vertex formula h = -b / (2a) with a = -3, b = 18.*
→ *h = -(18) / (2 * -3) = 3.*
→ *k = f(3) = -3(3)^2 + 18(3) + -33 = -6.*
→ *Vertex: (3, -6).*

Answer: $\text{Vertex} = (3,\ -6)$

14. Find the vertex of the parabola f(x) = 2x^2 + -4x + -4.

$$f(x) = 2x^2 - 4x - 4$$

→ *Use the vertex formula h = -b / (2a) with a = 2, b = -4.*
→ *h = -(-4) / (2 * 2) = 1.*
→ *k = f(1) = 2(1)^2 + -4(1) + -4 = -6.*
→ *Vertex: (1, -6).*

Answer: $\text{Vertex} = (1,\ -6)$

15. Find the vertex of the parabola f(x) = 2x^2 + -12x + 18.

$$f(x) = 2x^2 - 12x + 18$$

→ *Use the vertex formula h = -b / (2a) with a = 2, b = -12.*
→ *h = -(-12) / (2 * 2) = 3.*
→ *k = f(3) = 2(3)^2 + -12(3) + 18 = 0.*
→ *Vertex: (3, 0).*

Answer: $\text{Vertex} = (3,\ 0)$

16. Find the vertex of the parabola f(x) = 1x^2 + -10x + 30.

$$f(x) = 1x^2 - 10x + 30$$

→ *Use the vertex formula h = -b / (2a) with a = 1, b = -10.*
→ *h = -(-10) / (2 * 1) = 5.*
→ *k = f(5) = 1(5)^2 + -10(5) + 30 = 5.*
→ *Vertex: (5, 5).*

Answer: $\text{Vertex} = (5,\ 5)$

17. Find the vertex of the parabola f(x) = -1x^2 + 10x + -22.

$$f(x) = -1x^2 + 10x - 22$$

→ *Use the vertex formula h = -b / (2a) with a = -1, b = 10.*
→ *h = -(10) / (2 * -1) = 5.*
→ *k = f(5) = -1(5)^2 + 10(5) + -22 = 3.*
→ *Vertex: (5, 3).*

Answer: $\text{Vertex} = (5,\ 3)$

18. Find the vertex of the parabola f(x) = 1x^2 + 6x + 2.

$$f(x) = 1x^2 + 6x + 2$$

→ *Use the vertex formula h = -b / (2a) with a = 1, b = 6.*
→ *h = -(6) / (2 * 1) = -3.*
→ *k = f(-3) = 1(-3)^2 + 6(-3) + 2 = -7.*
→ *Vertex: (-3, -7).*

Answer: $\text{Vertex} = (-3,\ -7)$

Domain of a rational function

19. Find the value(s) excluded from the domain of f(x) = (3x + 1) / (4x + -1).

$$f(x) = \frac{3x + 1}{4x + -1}$$

→ *Set the denominator equal to zero: 4x + -1 = 0.*
→ *Solve: 4x = --1, so x = 1/4.*
→ *The domain is all real numbers except x = 1/4.*

Answer: $x \neq 1/4$

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20. Find the excluded value from the domain of g(x) = 2 / (x - 4).

$$g(x) = \frac{2}{x - 4}$$

→ *The denominator is x - 4. Set it equal to zero: x - 4 = 0.*
→ *Solve: x = 4.*
→ *Domain: all real numbers except x = 4.*

Answer: $x \neq 4$

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21. Find the value(s) excluded from the domain of f(x) = (4x + -1) / (3x + -2).

$$f(x) = \frac{4x + -1}{3x + -2}$$

→ *Set the denominator equal to zero: 3x + -2 = 0.*
→ *Solve: 3x = --2, so x = 2/3.*
→ *The domain is all real numbers except x = 2/3.*

Answer: $x \neq 2/3$

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22. Find the excluded value from the domain of g(x) = 4 / (x - 8).

$$g(x) = \frac{4}{x - 8}$$

→ *The denominator is x - 8. Set it equal to zero: x - 8 = 0.*
→ *Solve: x = 8.*
→ *Domain: all real numbers except x = 8.*

Answer: $x \neq 8$

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23. Find the value(s) excluded from the domain of f(x) = (1x + -2) / (2x + 0).

$$f(x) = \frac{1x + -2}{2x + 0}$$

→ *Set the denominator equal to zero: 2x + 0 = 0.*
→ *Solve: 2x = -0, so x = 0.*
→ *The domain is all real numbers except x = 0.*

Answer: $x \neq 0$

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24. Find the excluded value from the domain of g(x) = 9 / (x - -8).

$$g(x) = \frac{9}{x - -8}$$

→ *The denominator is x - -8. Set it equal to zero: x - -8 = 0.*
→ *Solve: x = -8.*
→ *Domain: all real numbers except x = -8.*

Answer: $x \neq -8$

25. Find the value(s) excluded from the domain of f(x) = (3x + -1) / (5x + -5).

$$f(x) = \frac{3x + -1}{5x + -5}$$

→ *Set the denominator equal to zero: 5x + -5 = 0.*
→ *Solve: 5x = --5, so x = 1.*
→ *The domain is all real numbers except x = 1.*

Answer: $x \neq 1$

26. Find the excluded value from the domain of g(x) = 8 / (x - -3).

$$g(x) = \frac{8}{x - -3}$$

→ *The denominator is x - -3. Set it equal to zero: x - -3 = 0.*
→ *Solve: x = -3.*
→ *Domain: all real numbers except x = -3.*

Answer: $x \neq -3$

27. Find the value(s) excluded from the domain of f(x) = (4x + -1) / (3x + 7).

$$f(x) = \frac{4x + -1}{3x + 7}$$

→ *Set the denominator equal to zero: 3x + 7 = 0.*
→ *Solve: 3x = -7, so x = -7/3.*
→ *The domain is all real numbers except x = -7/3.*

Answer: $x \neq -7/3$

28. Find the excluded value from the domain of g(x) = 2 / (x - 1).

$$g(x) = \frac{2}{x - 1}$$

→ *The denominator is x - 1. Set it equal to zero: x - 1 = 0.*
→ *Solve: x = 1.*
→ *Domain: all real numbers except x = 1.*

Answer: $x \neq 1$

29. Find the value(s) excluded from the domain of f(x) = (1x + -1) / (4x + 5).

$$f(x) = \frac{1x + -1}{4x + 5}$$

→ *Set the denominator equal to zero: 4x + 5 = 0.*
→ *Solve: 4x = -5, so x = -5/4.*
→ *The domain is all real numbers except x = -5/4.*

Answer: $x \neq -5/4$

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30. Find the excluded value from the domain of g(x) = 6 / (x - 8).

$$g(x) = \frac{6}{x - 8}$$

→ *The denominator is x - 8. Set it equal to zero: x - 8 = 0.*
→ *Solve: x = 8.*
→ *Domain: all real numbers except x = 8.*

Answer: $x \neq 8$
